

Clinical Digital Twin Report

Computational Cellular Reprogramming Analysis

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1. Introduction

The experiment aimed to explore the effects of cellular reprogramming and oncogenic stress on somatic cells, using a combination of OSKM transfection and p53 therapeutic intervention. This study is part of ongoing research to understand the dynamics of induced pluripotent stem cell (iPSC) generation and the potential for direct lineage conversion under stress conditions.

2. Materials and Methods

The experiment involved repeated transfection with OSKM factors, followed by oncogenic stress induction and p53 therapeutic intervention to modulate cell fate determination. Computational analyses were conducted using scVI, a single-cell variational inference tool, to assess the impact of these interventions on cellular populations.

Experimental Interventions:

- OSKM Transfection (Yamanaka)
- OSKM Transfection (Yamanaka)
- OSKM Transfection (Yamanaka)
- Oncogenic Stress Induced
- Oncogenic Stress Induced
- Oncogenic Stress Induced
- Oncogenic Stress Induced
- P53 Therapeutic Intervention
- P53 Therapeutic Intervention
- P53 Therapeutic Intervention
- Direct Neuronal Transdifferentiation

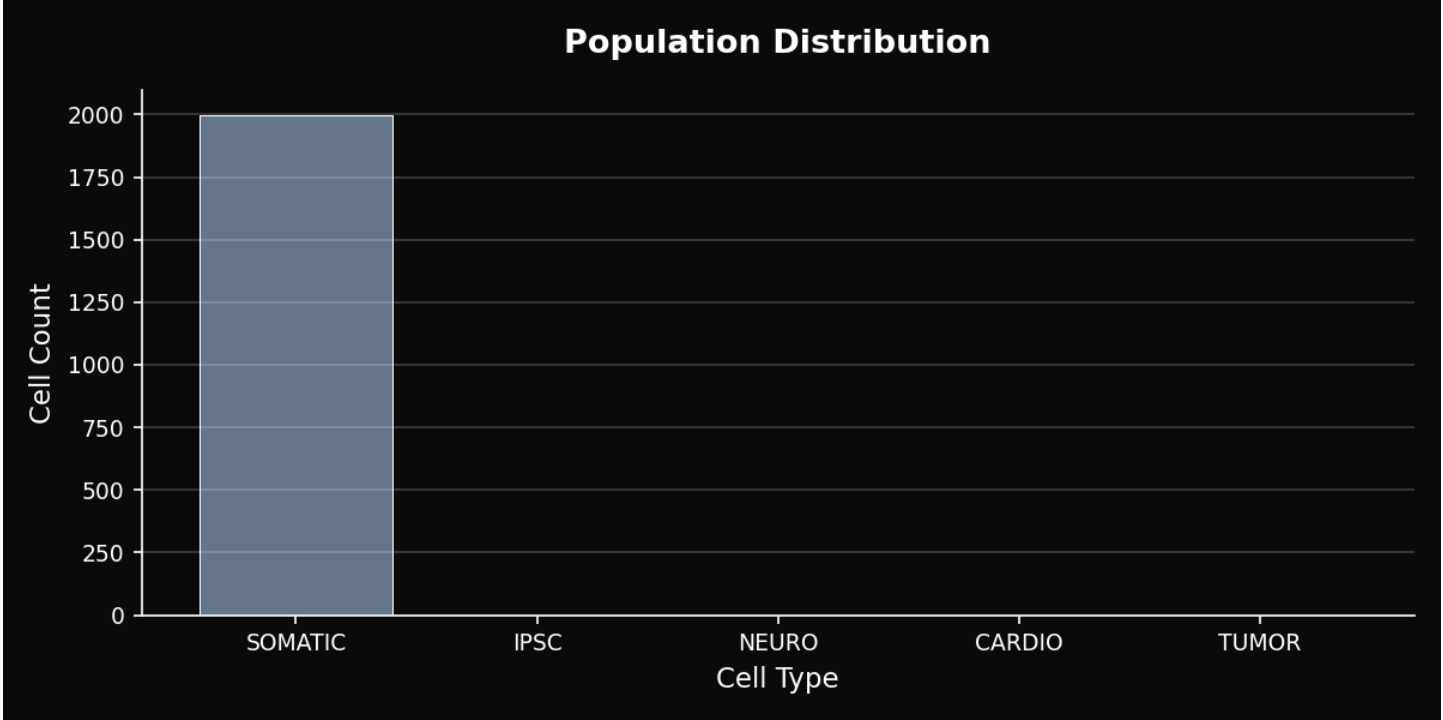
3. Results

Despite multiple rounds of OSKM transfection and oncogenic stress, the final cell population consisted entirely of somatic cells, with 1999 cells identified, and no detectable iPSC, neuronal, cardiomyocyte, or tumor cells. The scVI analyses revealed no significant reprogramming or transdifferentiation events. These findings suggest that the experimental conditions did not facilitate successful conversion or reprogramming of the somatic cells.

3.1 Population Distribution

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Cell Type	Count	Percentage
SOMATIC	1999	100.0%
IPSC	0	0.0%
NEURO	0	0.0%
CARDIO	0	0.0%
TUMOR	0	0.0%

3.2 Transcriptomic Analysis (scVI)

High-Resolution Gene Expression Analyses Performed: 4

Cells were expanded to high-dimensional transcriptomic space using a pre-trained variational autoencoder model on Human Cell Atlas data.

4. Discussion

The absence of iPSC or transdifferentiated cell populations indicates that the current protocol may be insufficient under the tested conditions, possibly due to inadequate reprogramming efficiency or the overpowering effects of oncogenic stress. The lack of tumorigenesis suggests effective p53-mediated tumor suppression but also highlights the challenge of balancing reprogramming with oncogenic control. Future studies should consider optimizing transfection efficiency and exploring alternative stress modulation strategies to enhance cellular plasticity and lineage conversion.

This report was automatically generated by the IS-CHRP v25.0 Clinical AI Bridge platform. The data represents computational simulation results from a high-fidelity cellular reprogramming model.